

Spherical friezes



Katie Waddle
University of Michigan

Plan for the talk

- ① Coxeter-Conway friezes: algebraic construction
- ② Coxeter-Conway friezes: geometric interpretations
- ③ Euclidean distance geometry: Heronian friezes
- ④ Spherical distance geometry: Spherical Heronian friezes

K. Waddle, *Spherical friezes*, arXiv:2501.03587, 2025.

An algebraic construction

Definition [Coxeter, 1971, Conway and Coxeter, 1973]

A *frieze pattern* is a multi-line array of positive real numbers bordered by two rows of 1's, such that for each 2×2 block $\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}$, we have $ac - bd = 1$.

Survey: [Morier-Genoud, 2015]

$$\begin{array}{cccccccc} \cdots & 1 & 1 & 1 & 1 & 1 & 1 & \\ & x_1 & & & & & & \cdots \\ \cdots & x_2 & & & & & & \\ & 1 & 1 & 1 & 1 & 1 & 1 & \cdots \end{array}$$

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Survey: [Morier-Genoud, 2015]

The diagram shows a frieze pattern with three rows of numbers. The top row consists of 1's. The bottom row also consists of 1's. The middle row contains algebraic expressions. A central parallelogram is shaded light gray, containing the following expressions:

$\dots$	1	1	1	1	1	1	$\dots$
x_1	$\frac{x_2+1}{x_1}$	$\frac{x_1+1}{x_2}$	x_2	$\frac{1+x_2+x_1}{x_2x_1}$	x_1	$\dots$	
$\dots$	x_2	$\frac{x_1+x_2+1}{x_1x_2}$	x_1	$\frac{1+x_2}{x_1}$	$\frac{1+x_1}{x_2}$	x_2	
1	1	1	1	1	1	$\dots$	

periodicity and glide symmetry

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Survey: [Morier-Genoud, 2015]

The diagram shows a frieze pattern with four rows of numbers. The top and bottom rows consist of 1's. The second and third rows contain variables and their reciprocals. A shaded gray triangle covers the left portion of the pattern. A specific entry in the second row, $\frac{1+x_2+x_1}{x_2x_1}$, is circled.

...	1	1	1	1	1	1	...
	x_1	$\frac{x_2+1}{x_1}$	$\frac{x_1+1}{x_2}$	x_2	$\frac{1+x_2+x_1}{x_2x_1}$	x_1	...
...		x_2	$\frac{x_1+x_2+1}{x_1x_2}$	x_1	$\frac{1+x_2}{x_1}$	$\frac{1+x_1}{x_2}$	x_2
	1	1	1	1	1	1	...

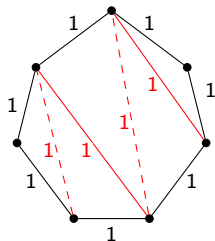
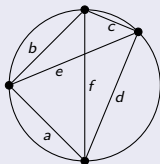
Laurent phenomenon

periodicity and glide symmetry

A combinatorial interpretation

Theorem [Ptolemy, ≈ 150 CE]

Consider a cyclic quadrilateral with consecutive sides a, b, c, d and diagonals e, f . Then $ac + bd = ef$.

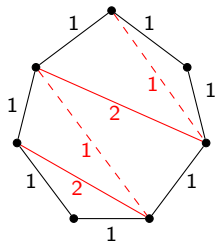
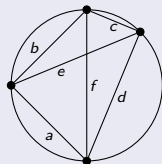


	1	1	1	1	1	1	1	1
1	2	3	2	1	3	3	1	
	1	5	5	1	2	8	2	1
1	2	8	2	1	5	5	1	
	1	3	3	1	2	3	2	1
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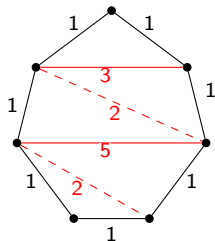
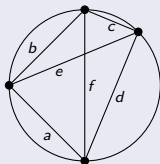


	1	1	1	1	1	1	1	1
1	2	3	2	1	3	3	1	
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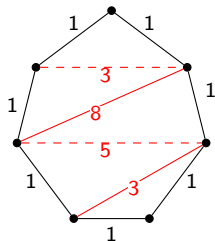
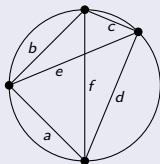


	1	1	1	1	1	1	1	1	1
1	2	3	2	1	3	3	1		
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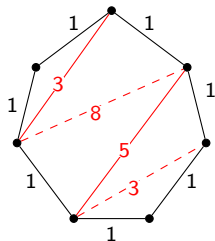
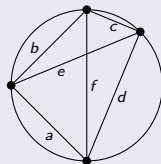


	1	1	1	1	1	1	1	1
1	2	3	2	1	3	3	1	
	1	5	5	1	2	8	2	1
1	2	8	2	1	5	5	1	
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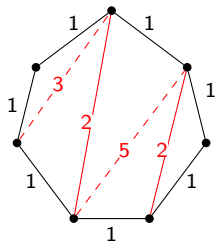
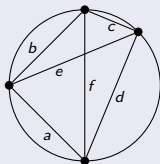


	1	1	1	1	1	1	1	1	1
1	2	3	2	1	3	3	1		
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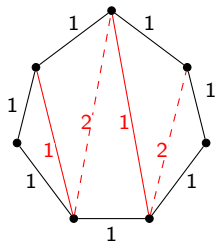
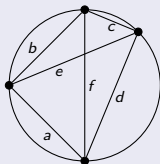


	1	1	1	1	1	1	1	1	1
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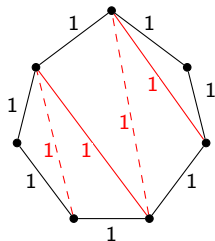
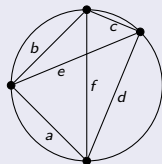


	1	1	1	1	1	1	1	1	1
1		2	3	2	1	3	3	1	
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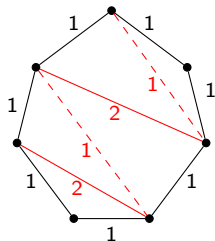
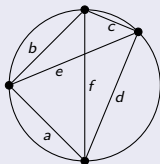


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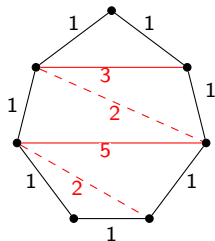
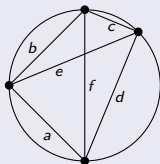


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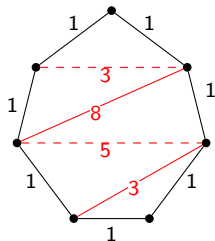
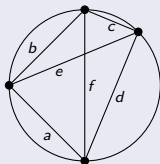


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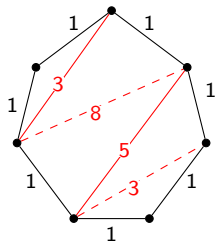
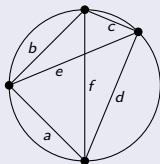


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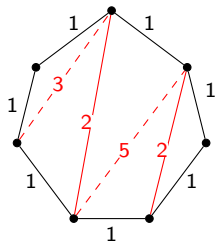
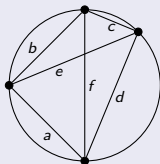


	1	1	1	1	1	1	1	1	1
1	2	3	2	1	3	1	3	1	1
1	1	5	5	1	2	8	2	1	1
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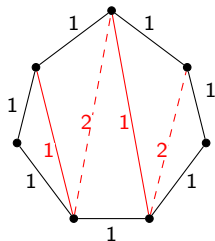
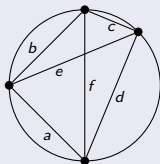


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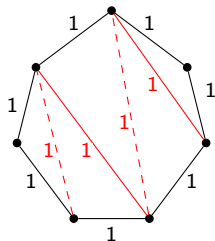
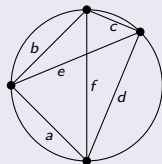


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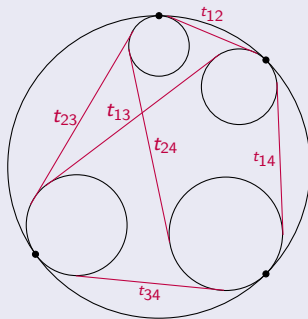


	1	1	1	1	1	1	1	1	1
1		2	3	2	1	3	3	1	1
	1		5	5	1	2	8	2	1
1		2		8	2	1	5	5	1
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1		1		1		1	1	1	1

Casey's Theorem

Theorem [Casey, 1866]

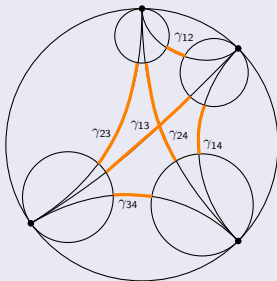
$$t_{12}t_{34} + t_{14}t_{23} = t_{13}t_{24}$$



Penner's lambda lengths

Theorem [Penner, 1987]

$$\lambda_{12}\lambda_{34} + \lambda_{14}\lambda_{23} = \lambda_{13}\lambda_{24}$$



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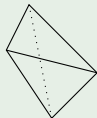
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A triangulation of an n -gon has $2n - 3$ pairwise distances.

Observation

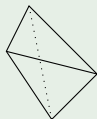
Distances corresponding to a triangulation are not sufficient to reconstruct a configuration up to oriented isometry.



Recovering a configuration from a small set of measurements

Observation

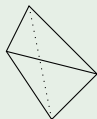
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Recovering a configuration from a small set of measurements

Observation

Distances corresponding to a triangulation are not sufficient to reconstruct a configuration up to oriented isometry.



Question

How can we supplement the data corresponding to a triangulation to enable unique reconstruction of an n -point configuration, up to oriented isometry?

The Cayley-Menger determinant

Definition

Given points $A_1, \dots, A_n$ in a Euclidean space $\mathbb{R}^d$, let x_{ij} denote the squared distance between A_i and A_j . The associated Cayley-Menger determinant is defined by

$$M_n^0(x_{12}, x_{13}, \dots, x_{n-1,n}) = \det \begin{bmatrix} 0 & 1 & 1 & \cdots & 1 & 1 \\ 1 & 0 & x_{12} & \cdots & x_{1,n-1} & x_{1,n} \\ 1 & x_{12} & 0 & & \vdots & \vdots \\ \vdots & \vdots & & \ddots & \vdots & \vdots \\ 1 & x_{1,n-1} & \cdots & \cdots & 0 & x_{n-1,n} \\ 1 & x_{1,n} & \cdots & \cdots & x_{n,n-1} & 0 \end{bmatrix}.$$

The (squared) volume of a simplex

Theorem

Consider a simplex $A_1 \cdots A_{d+1}$ in $\mathbb{R}^d$, with volume V . Let x_{ij} denote the squared distance from A_i to A_j . Then

$$V^2 = \frac{(-1)^{d+1}}{(d!)^2 \cdot 2^d} M_{d+1}^0(x_{12}, x_{13}, \dots, x_{23}, x_{24}, \dots, x_{d,d+1}).$$

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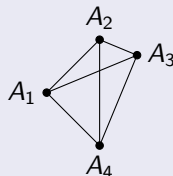
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Corollary

Given $A_1, A_2, A_3, A_4 \in \mathbb{R}^2$, we have

$$M_4^0(x_{12}, x_{13}, x_{14}, x_{23}, x_{24}, x_{34}) = 0.$$



Heron's formula

We will use the notation

$$\begin{aligned} H(a, b, c) &= -M_3^0(a, b, c) \\ &= -\det \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & a & b \\ 1 & a & 0 & c \\ 1 & b & c & 0 \end{bmatrix} \\ &= -a^2 - b^2 - c^2 + 2ab + 2ac + 2bc. \end{aligned}$$

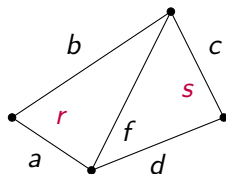
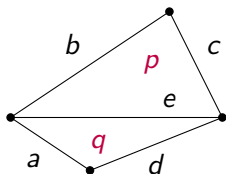
Theorem [Heron, ≈ 250 CE]

Let $A_1A_2A_3$ be a triangle with squared side lengths a, b, c . Then

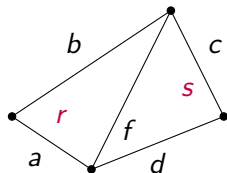
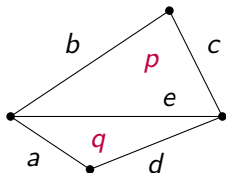
$$\underbrace{(4 \cdot \text{area of } A_1A_2A_3)}_S^2 = H(a, b, c).$$

The converse is also true if we work over $\mathbb{C}$.

Solution of the reconstruction problem on the Euclidean plane [Fomin-Setiabrata, 2021]

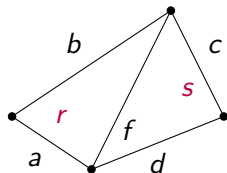
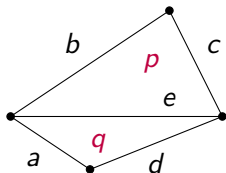


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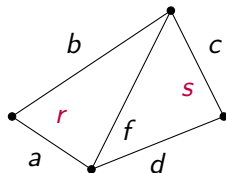
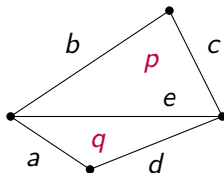
- $p^2 = H(b, c, e)$
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Solution of the reconstruction problem on the Euclidean plane [Fomin-Setiabrata, 2021]



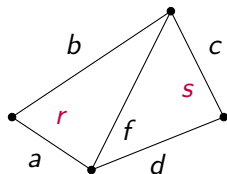
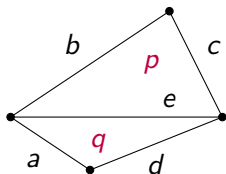
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Distance geometry on a sphere

Let $\mathbf{S} \subset \mathbb{R}^3$ be a sphere with radius R centered at the origin. Set $K = \frac{1}{R^2}$.

Problem

Identify a collection of $O(n)$ measurements that uniquely determines a labeled set of n points on $\mathbf{S}$, considered up to oriented isometry. Write the corresponding formulas explicitly.

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This rules out the use of geodesic distances, spherical areas, etc.

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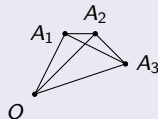
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Definition

$$S^K(A_1, A_2, A_3) = \frac{12}{R} V(OA_1A_2A_3)$$



Spherical Cayley-Menger determinant [T. Tao, 2019]

As before, $K = \frac{1}{R^2}$. We will use the notation

$$\begin{aligned} H^K(a, b, c) &= \frac{K}{2} \det \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & a & c & \frac{1}{K} \\ 1 & a & 0 & b & \frac{1}{K} \\ 1 & c & b & 0 & \frac{1}{K} \\ 1 & \frac{1}{K} & \frac{1}{K} & \frac{1}{K} & 0 \end{bmatrix} \\ &= -a^2 - b^2 - c^2 + 2ab + 2ac + 2bc - Kabc. \end{aligned}$$

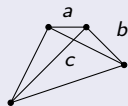
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Spherical Heron's formula

$$(S^K)^2 = H^K(a, b, c).$$

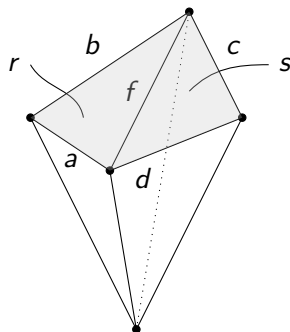
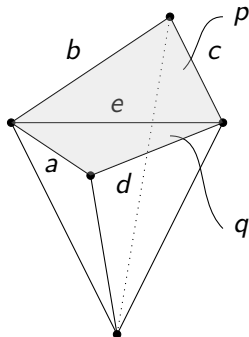


The converse holds if we work over $\mathbb{C}$.

Reconstruction problem on the sphere

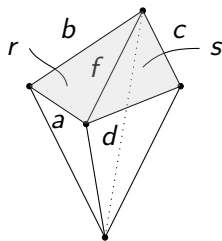
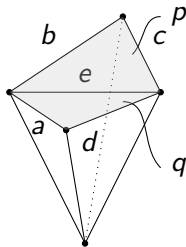
Problem

Given measurement data corresponding to a triangulation of a polygon (squared side distances and S^K measurements), compute all remaining distance measurements.

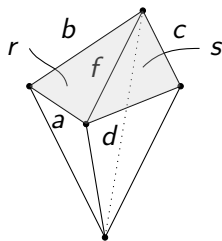
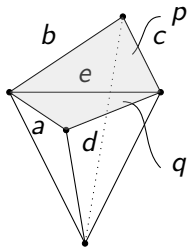


Solution on the sphere [W., 2025]

- $p^2 = H^K(b, c, e)$
- $q^2 = H^K(a, d, e)$
- $r^2 = H^K(a, f, b)$
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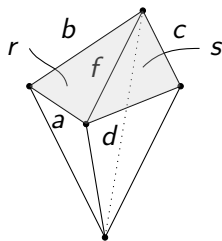
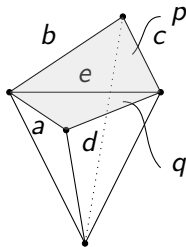


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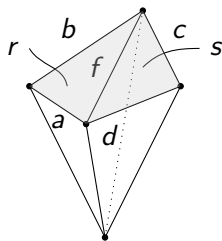
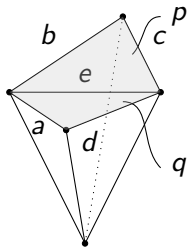
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- $s^2 = H^K(c, f, d)$
- $p + q = r + s + \frac{K}{2}(ap + bq - er)$
- $p + q = r + s + \frac{K}{2}(fp - cr - bs)$
- $p + q = r + s + \frac{K}{2}(dp + cq - es)$
- $p + q = r + s + \frac{K}{2}(fq - dr - as)$

Solution on the sphere [W., 2025]



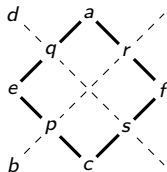
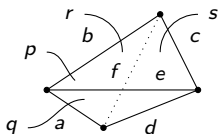
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Spherical Heronian diamond

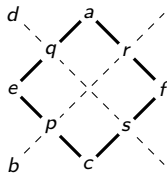
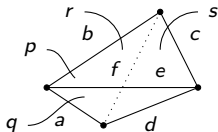


Definition

A *spherical Heronian diamond* is a 10-tuple $(a, b, c, d, e, f, p, q, r, s)$ satisfying the equations

- (1) $p^2 = H^K(b, c, e)$ $q^2 = H^K(a, d, e)$
- (2) $r^2 = H^K(a, f, b)$ $s^2 = H^K(c, f, d)$
- (3) $p + q = r + s + \frac{K}{2}(ap + bq - er)$
- (4) $e(r - s) = p(a - d) + q(b - c)$
- (5) $4ef = (p + q)^2 + (a - b + c - d)^2 - Ke(a - b)(c - d)$

Spherical Heronian diamond: propagation rule



Proposition [W., 2025]

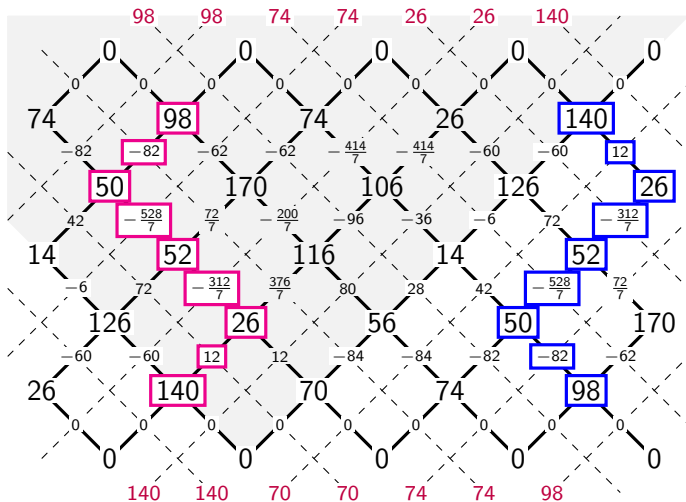
Suppose that a, b, c, d, e, p, q satisfy spherical Heronian relations (1). Assuming $e \notin \{0, \frac{4}{K}\}$, there exist unique f, r, s such that $(a, b, c, d, e, f, p, q, r, s)$ is a spherical Heronian diamond. Namely,

$$f = \frac{(p+q)^2 + (a-b+c-d)^2 - Ke(a-b)(c-d)}{4e\left(1 - \frac{Ke}{4}\right)},$$

$$r = \frac{p\left(e+a-d - \frac{Kae}{2}\right) + q\left(e-c+b - \frac{Kbe}{2}\right)}{2e\left(1 - \frac{Ke}{4}\right)},$$

$$s = \frac{p\left(e-a+d - \frac{Kde}{2}\right) + q\left(e+c-b - \frac{Kce}{2}\right)}{2e\left(1 - \frac{Ke}{4}\right)}.$$

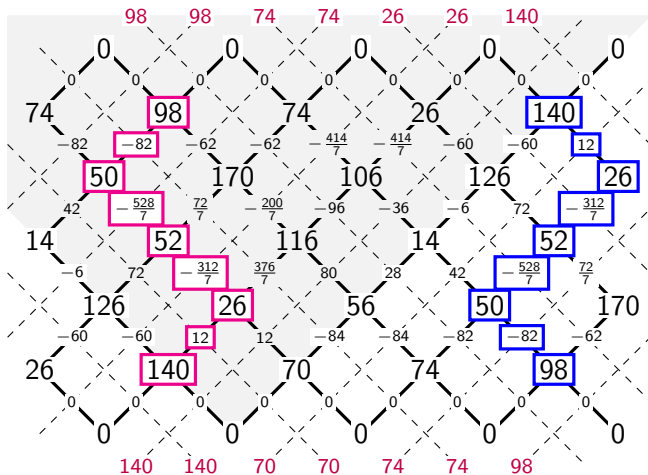
Spherical Heronian friezes



$$K = \frac{1}{49}$$

Main result

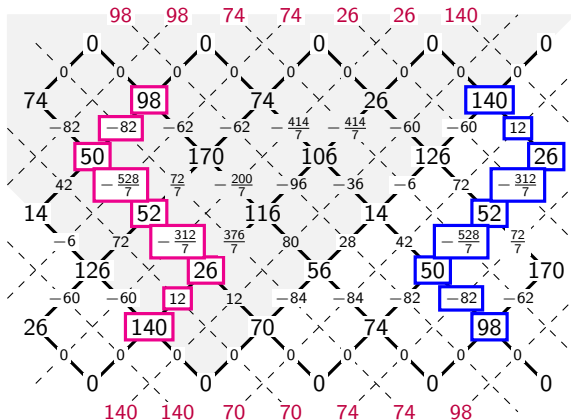
The entries along a **traversing path** determine the entire frieze.



Main result

Theorem [W., 2025]

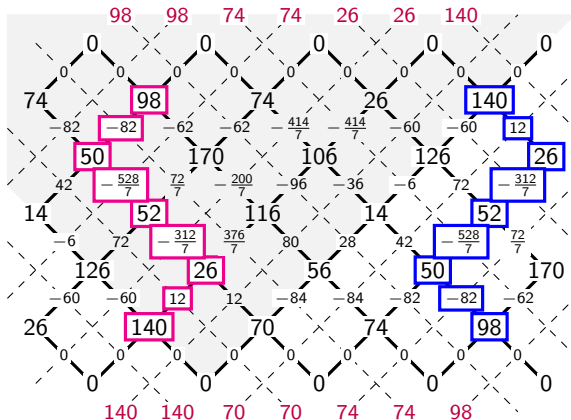
A sufficiently generic spherical Heronian frieze is periodic, has glide symmetry, and exhibits a form of the Laurent phenomenon.



Main result

Theorem [W., 2025]

Every generic spherical Heronian frieze comes from a polygon on the complexified sphere. This polygon is unique.



Thank you!

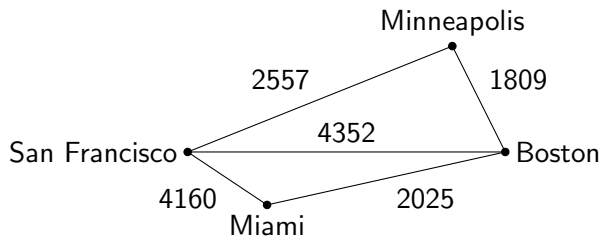


References

-  J. H. Conway and H. S. M Coxeter, *Triangulated polygons and frieze patterns*, Math. Gaz. **57** (1973), no. 400, 87–94, no. 401, 87–94.
-  H. S. M. Coxeter, *Frieze patterns*, Acta Arith. **18** (1971), no. 1, 297–310.
-  S. Fomin and L. Setiabrata, *Heronian friezes*, Int. Math. Res. Not. **2021** (2021), no. 1, 648–694.
-  S. Morier-Genoud, *Coxeter's frieze patterns at the crossroads of algebra, geometry and combinatorics*, Bull. Lond. Math. Soc. **47** (2015), no. 6, 895–938.
-  T. Tao, *The spherical Cayley-Menger determinant and the radius of the Earth*, 2019.

What next?

One application of spherical Heronian friezes is to measuring and computing distances on a globe.



Question

What if we have a point (or points) on a different sphere, like a satellite?